2.2.14

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Question:
The angle between the line

$$\mathbf{r} = \left(5\hat{i} - \hat{j} - 4\hat{k}\right) + \lambda\left(2\hat{i} - \hat{j} + \hat{k}\right) \tag{0.1}$$

and the plane

$$\mathbf{r}.\left(3\hat{i} - 4\hat{j} - \hat{k}\right) + 5 = 0 \tag{0.2}$$

is
Solution: The given line can be expressed in the form as

$$\mathbf{r} = \begin{pmatrix} 5 \\ -1 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \tag{0.3}$$

Hence the vector direction of this line is

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \tag{0.4}$$

the normal vector of the given plane is

$$\begin{pmatrix} 3 \\ -4 \\ -1 \end{pmatrix} \tag{0.5}$$

by the formula

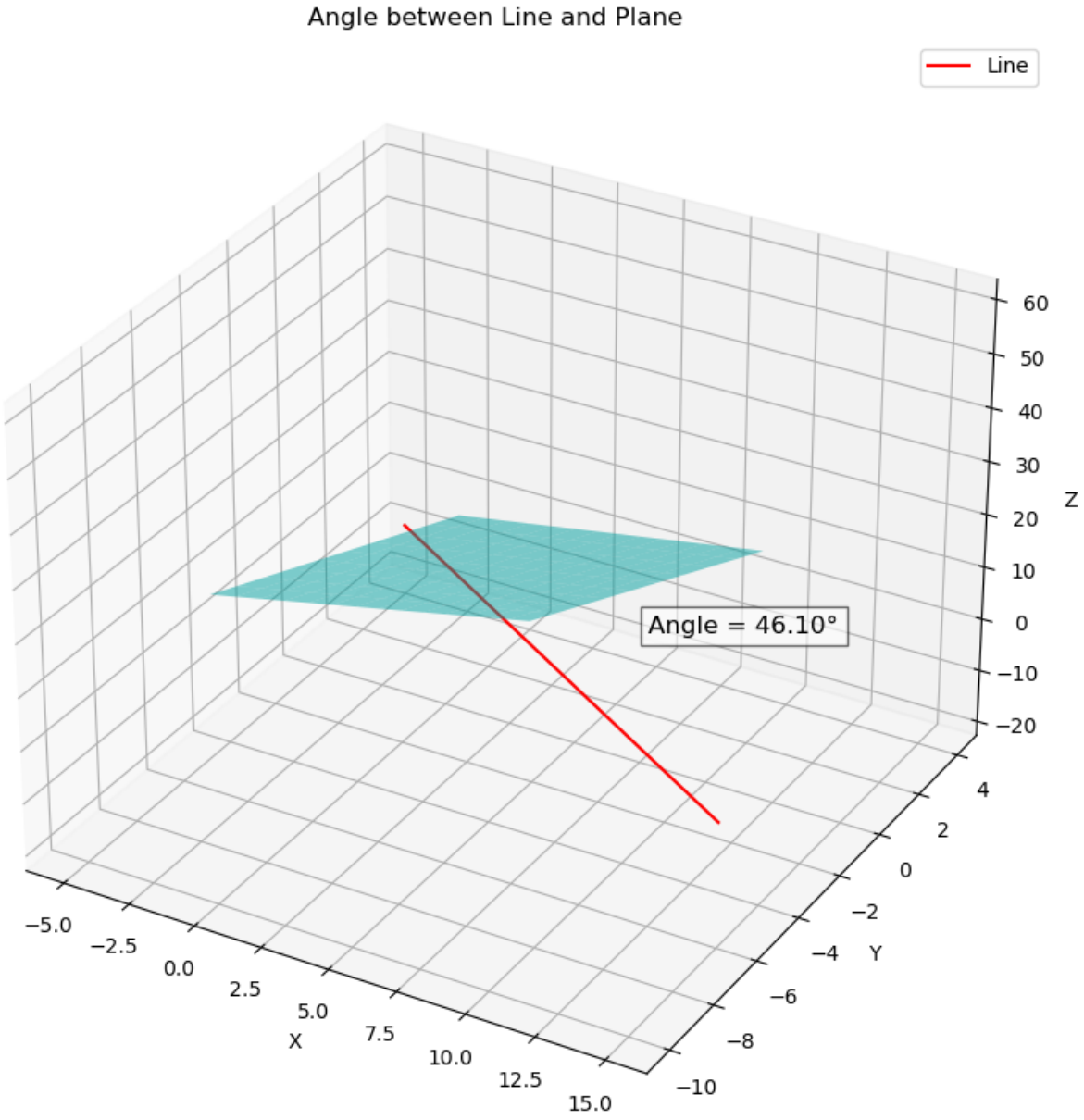
$$cos\theta = \frac{\mathbf{a}^T\mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} \tag{0.6}$$

Thus the cosine of the angle between the two is

$$\frac{3\sqrt{3}}{2\sqrt{13}} \tag{0.7}$$

which is sine of the angle between the plane and the line.

$\therefore$ The angle between the line and plane is $sin^{-}1\frac{3\sqrt{3}}{2\sqrt{13}}$


Angle between Line and Plane
Line
Angle = 46.10°
X
Y
Z
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
−10
−8
−6
−4
−2
0
2
4
−20
−10
0
10
20
30
40
50
60